

1. A particle of mass m moves in the xy -plane. Initially, $t = 0$, its position and velocity are

$$\vec{r}(0) = x_0 \hat{x}, \quad \vec{v}(0) = v_0 \hat{y}$$

For $t > 0$, it is acted upon by a constant force, where F_0 is a positive constant.

$$\vec{F} = F_0(2\hat{x} - \hat{y})$$

(a.) Use Newton's second Law to find the acceleration $\vec{a}(t)$ or $\ddot{\vec{r}}(t)$.

$$\Sigma F = ma = m\ddot{\vec{r}}, \quad \ddot{\vec{r}}(t) = \frac{F_0}{m}(2\hat{x} - \hat{y})$$

(b.) Find the velocity $\vec{v}(t)$ or $\dot{\vec{r}}(t)$.

$$\vec{v}(t) = \dot{\vec{r}}(t) = \int \ddot{\vec{r}}(t) dt = \frac{F_0}{m} \int (2\hat{x} - \hat{y}) dt$$

$$\dot{\vec{r}}(t) = \frac{F_0}{m}(2t\hat{x} - t\hat{y}) + C, \quad \dot{\vec{r}}(t=0) = C = v_0 \hat{y}$$

$$\therefore \dot{\vec{r}}(t) = \frac{2F_0}{m}t\hat{x} + (v_0 - \frac{F_0}{m}t)\hat{y}$$

(c.) Find the position $\vec{r}(t)$.

$$\begin{aligned} \vec{r}(t) &= \int \dot{\vec{r}}(t) dt = \int \left(\frac{2F_0}{m}t\hat{x} + (v_0 - \frac{F_0}{m}t)\hat{y} \right) dt \\ &= \frac{F_0}{m}t^2\hat{x} + (v_0t - \frac{F_0}{2m}t^2)\hat{y} + C, \quad \vec{r}(t=0) = C = x_0\hat{x} \end{aligned}$$

$$\therefore \vec{r}(t) = \left(x_0 + \frac{F_0}{m}t^2 \right) \hat{x} + \left(v_0t - \frac{F_0}{2m}t^2 \right) \hat{y}$$

(d.) At what time is the y -component of the particle's velocity equal to zero?

$$\dot{r}_y(t) = v_0 - \frac{F_0}{m}t = 0, \quad t = \frac{v_0 m}{F_0}$$

2. A particle of mass m moves in a circle of **constant radius** R . Its angular position is:

$$\phi(t) = \omega_0 t + \frac{1}{2} \alpha t^2$$

where ω_0 and α are constants.

(a.) Determine $\dot{\mathbf{r}}$, $\ddot{\mathbf{r}}$, $\dot{\phi}$ and $\ddot{\phi}$.

$$r = R, \quad \dot{r} = 0, \quad \ddot{r} = 0$$

$$\dot{\phi} = \omega_0 + \alpha t$$

$$\ddot{\phi} = \alpha$$

(b.) Using the polar-coordinate expression for acceleration:

$$\vec{a} = (\ddot{r} - r\dot{\phi}^2)\hat{\mathbf{r}} + (r\ddot{\phi} + 2\dot{r}\dot{\phi})\hat{\phi}$$

find $\vec{a}(t)$.

$$\vec{a} = (-r\dot{\phi}^2)\hat{\mathbf{r}} + (r\ddot{\phi})\hat{\phi}$$

$$= -R(\omega_0 + \alpha t)^2 \hat{\mathbf{r}} + R\alpha \hat{\phi}$$

(c.) Use Newton's second law to find the radial and tangential components of the net force.

$$\sum \vec{F} = m\vec{a}, \quad F_{\hat{\mathbf{r}}} = -mR(\omega_0 + \alpha t)^2 \hat{\mathbf{r}}$$

$$F_{\hat{\phi}} = mR\alpha$$

(d.) Briefly comment on the physical role of each force component.

$F_{\hat{\mathbf{r}}}$ is a radially inward force providing centripetal acceleration

$F_{\hat{\phi}}$ is a tangential force providing an angular acceleration